Construction

In the gear box, lay shaft or counter shaft is present. This shaft is supported by the bearings in the gearbox. The gear teeth are fixed as stationary in this lay shaft. In the main shaft, splines are made in such a way that the gears are free to move on the slots. The gear teethed wheel (1) in the clutch shaft is always in mesh with the gear in the lay shaft wheel (2). The gear (7) in the lay shaft is always in mesh with the idler gear (8). A three speed sliding mesh gear box is shown in Figure 3.5.1.

Working principle

a) Neutral position

The gear teeth (1) in the input (clutch) shaft is in mesh with the counter shaft gear 2. Hence the counter shaft rotates. However the gears (3 or 4) are not in mesh with the output shaft gears (5 or 6) as shown in Figure 3.5.1.1. Hence the power will not be transmitted to the output shaft. This condition is called as neutral position.

b) First gear

The gear (1) in the clutch shaft is in mesh with the gear in the counter shaft (2). As the gears 3 and 4 are in the counter shaft they also rotate. When the gear shift fork is moved towards left as shown in Figure 3.5.1.1 to engage the gear 5 in the output shaft with the gear 4 in the lay shaft, then the output shaft rotates. The rotary power is now transmitted to the output shaft from gear 1-2-4-5. Now we can get first gear ratio 3:1.

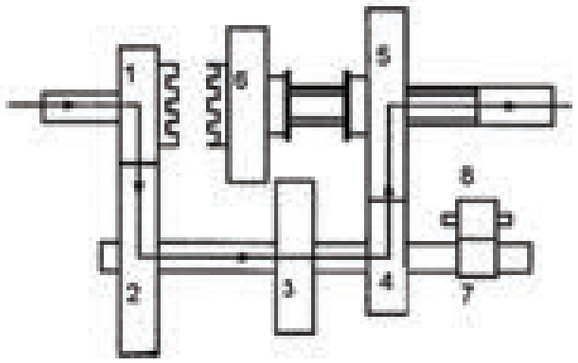


Figure 3.5.1.1 First gear power transmission

c) Second gear

During second gear, the gear shift fork is moved right as shown in Figure 3.5.1.2 to mesh the gear 6 in the output shaft with gear 3 in the lay shaft gear 1, 2, 3 and 6 will get contact and rotate. Now the power is transmitted from the gear 1-2-3-6. Now the speed becomes more than the first gear and the gear ratio of 2:1 is achieved.

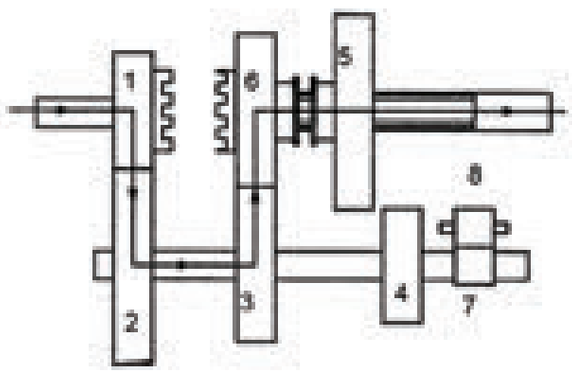


Figure 3.5.1.2 Second gear power transmission

d) Third gear

Dog Clutch is located on the gear 1 in the input (clutch) shaft and the gear 6 in the output shaft as shown in

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